

CO1-AT1

1. What is Data Science? Explain its primary objective.

Data Science is an interdisciplinary field that uses statistics, machine learning, programming, and domain knowledge to extract meaningful insights from data.

Primary Objective:

- To analyse data and discover useful patterns.
- To support better decision-making.
- To predict future trends.
- To solve real-world problems using data.

2. Why is Data Science important in today's digital world?

Mention any four reasons.

1. Helps organizations make data-driven decisions.
2. Predicts future trends and customer behaviour.
3. Improves business efficiency and productivity.
4. Enables innovation through AI and machine learning.

3. List any four benefits of Data Science in business and society.

- Better decision-making.
- Improved customer experience.
- Fraud detection and risk management.
- Healthcare diagnosis and disease prediction.

4. Mention four real-world applications of Data Science.

- Healthcare (Disease Prediction)
- Banking (Fraud Detection)

- E-commerce (Product Recommendation)
- Transportation (Traffic Prediction)

5. What are the different facets of data? Briefly explain each.

- **Structured Data:** Organized in rows and columns (e.g., Database tables).
- **Semi-Structured Data:** Contains tags but no fixed format (e.g., JSON, XML).
- **Unstructured Data:** No predefined format (e.g., Images, Videos, Emails).

6. Differentiate between structured, semi-structured, and unstructured data with examples.

Structured	Semi-Structured	Unstructured
Fixed format	Partial structure	No structure
Stored in databases	JSON, XML	Images, Videos
Easy to analyse	Moderately easy	Difficult to analyse

7. What is Big Data? State its key characteristics (5 Vs).

Big Data refers to extremely large datasets that cannot be processed using traditional tools.

5 Vs:

- **Volume** – Large amount of data.
- **Velocity** – Speed of data generation.
- **Variety** – Different data formats.
- **Veracity** – Data quality and accuracy.
- **Value** – Useful insights obtained from data.

8. Explain the Big Data ecosystem. Name any four technologies used in it.

The Big Data ecosystem consists of tools used to collect, store, process, analyse, and visualize large datasets.

Technologies:

- Hadoop
- Apache Spark
- Hive
- HBase

9. What are the major phases of the Data Science process?

1. Data Collection
2. Data Cleaning
3. Data Integration
4. Data Transformation
5. Data Analysis
6. Model Building
7. Model Evaluation
8. Deployment
9. Presentation of Results

10. What is data retrieval? Mention two common sources from which data can be retrieved.

Data retrieval is the process of collecting data from different sources for analysis.

Sources: Databases, Websites/APIs

11. Why is data collection considered an important step in the Data Science process?

- Provides quality data for analysis.
- Improves model accuracy.
- Reduces errors.
- Supports better decision-making.

12. What is data cleansing? List any four common data-cleaning techniques.

Data cleansing is the process of improving data quality by removing errors and inconsistencies.

Techniques:

- Remove duplicate records.
- Handle missing values.
- Correct inconsistent data.
- Remove outliers.

13. What is data integration? Why is it necessary in Data Science?

Data integration combines data from multiple sources into a single dataset.

Importance:

- Provides complete information.
- Improves data consistency.
- Reduces redundancy.
- Enhances analysis quality.

14. Explain data transformation with suitable examples.

Data transformation converts data into a suitable format for analysis.

Examples:

- Converting text labels into numbers.
 - Normalizing numerical values.
 - Changing date formats.
 - Scaling features.
-

15. What is data analysis? Mention any four techniques used for data analysis.

Data analysis is the process of examining data to discover useful information.

Techniques:

- Statistical Analysis
- Data Visualization
- Machine Learning
- Data Mining

16. Differentiate between descriptive, predictive, and prescriptive analytics.

Descriptive	Predictive	Prescriptive
What happened?	What will happen?	What should be done?
Uses historical data	Uses ML models	Recommends actions
Example: Sales report	Sales forecasting	Inventory optimization

17. What is meant by building a model in Data Science? Mention the basic steps involved.

Model building is creating a machine learning model to make predictions or classifications.

Steps:

1. Select algorithm.
2. Split data into training and testing sets.
3. Train the model.
4. Test the model.
5. Improve model performance.

18. Why is model evaluation important? Name any four-evaluation metrics used for classification models.

Model evaluation measures how accurately a model performs.

Metrics:

- Accuracy
- Precision
- Recall
- F1-Score

19. Why is presenting findings an essential part of the Data Science process? Mention any four visualization tools or techniques.

Presenting findings helps stakeholders understand insights and make informed decisions.

Visualization Tools/Techniques:

- Bar Chart

- Pie Chart
- Line Chart
- Tableau (or Microsoft Power BI)

20. How are Data Science models deployed into real-world applications? Give any four examples of Data Science applications.

After testing, models are deployed as web applications, mobile apps, cloud services, or APIs so users can make predictions in real time.

Applications:

- Fraud Detection in Banking
- Disease Prediction in Healthcare
- Recommendation Systems (Amazon, Netflix)
- Spam Email Detection